WORKED SOLUTION

STEP III 2007, Question 11

Mechanics • Friction

STEP III 2007, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q11

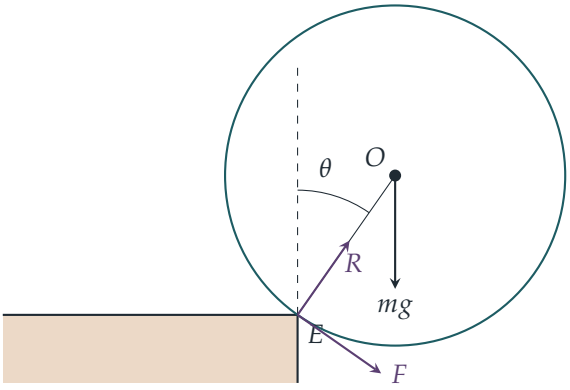
The key idea

Once the rim reaches the edge, the edge acts as a pivot: the point of the rim touching it is at rest, so the centre moves on a circle of radius a about the edge. We split the contact force at the edge into a component R along the radius (the normal reaction) and a component F along the rim (friction), and compare them with the motion of the centre of mass. The whole question turns on where the mass sits. In (i) all of it is at the centre, so the wheel has no moment of inertia about its centre. In (ii) the mass is out on the rim, and that is what produces a friction force.

Solution

Model and notation. Rim, spokes and hoop are light and rigid; the hub and the two particles are particles. Let E be the point of the edge in contact with the rim, O the centre, and θ the angle EO makes with the upward vertical, so $\theta = 0$ at the instant the wheel arrives at the edge. While the rim stays in contact with E without slipping, the material point of the rim at E is at rest. So the rigid body rotates about E with angular speed $\dot{\theta}$, and O moves on a circle of radius a about E with speed $a\dot{\theta}$.

The contact force at E has a component R along EO (away from E) and a component F perpendicular to EO, taken positive in the direction of increasing θ. Because E is at rest, the contact force does no work, so energy is conserved.



Here R acts along EO, and F is drawn in its positive sense.

For the centre of mass G at distance a from E, its acceleration has components $a\ddot{\theta}$ along the direction of motion and $a\dot{\theta}^2$ towards E. Resolving the external forces on the whole body (gravity mg and the contact force):

$$\text{along the motion:} \quad ma\ddot{\theta} = mg\sin\theta + F, \tag{1}$$

$$\text{towards } E: \quad ma\dot{\theta}^2 = mg\cos\theta - R. \tag{2}$$

In both parts the centre of mass is at O (the hub; the midpoint of the two particles), so (1) and (2) hold in both.

Part (i)

On the table the wheel moves at constant speed u, so at $\theta = 0$ we have $\dot{\theta} = u/a$. The only kinetic energy is that of the hub, $\frac{1}{2}ma^2\dot{\theta}^2$, and the hub has dropped $a(1-\cos\theta)$, so

$$a^2\dot{\theta}^2 = u^2 + 2ga(1-\cos\theta).$$

Differentiating with respect to time and cancelling $2a\dot{\theta}$ (which is non-zero, since $\dot{\theta} \geq u/a > 0$):

$$a\ddot{\theta} = g\sin\theta.$$

Substituting in (1) gives $F = 0$: **the frictional force is zero**, as required.

There is also a direct reason. The rim and spokes are light, so the total moment about O of the forces on them must vanish. Those forces are the contact force at E and the force from the hub at O, which has no moment about O. The moment of the contact force about O is Fa, so $F = 0$.

From (2), the normal reaction is

$$R = mg\cos\theta - \frac{m}{a}(u^2 + 2ga(1-\cos\theta)) = mg(3\cos\theta - 2) - \frac{mu^2}{a}.$$

At $\theta = 0$ this is $m(g - u^2/a) > 0$, because $u^2 < ag$, so the wheel does not leave the edge at once. R then decreases as θ increases, and the rim loses contact when $R = 0$, that is

when

$$\cos\theta = \frac{2ag + u^2}{3ag},$$

which is less than 1 because $u^2 < ag$. The hub has then fallen

$$a(1 - \cos\theta) = a \cdot \frac{3ag - 2ag - u^2}{3ag} = \frac{ag - u^2}{3g},$$

as required. Since $F = 0$ throughout, the no-slip assumption needs no help from friction, and the analysis holds right up to this point.

Part (ii)

? **Ask yourself** Where are the two particles while the hoop pivots about E?

At $\theta = 0$ the diameter through the particles is vertical, so the lower particle is at the contact point E. The hoop then rotates rigidly about the material point at E, which is that particle, so the particle stays at E. The other particle stays diametrically opposite, at distance $2a$ from E along EO.

So the moment of inertia about E is $\frac{m}{2}(2a)^2 = 2ma^2$. On the table the hoop rolls at constant speed u (nothing does work), so $\dot\theta = u/a$ at $\theta = 0$. The moving particle is at height $2a\cos\theta$ above E, and energy conservation gives

$$\tfrac{1}{2}(2ma^2)\dot\theta^2 + \tfrac{m}{2}g(2a\cos\theta) = \tfrac{1}{2}(2ma^2)\frac{u^2}{a^2} + \tfrac{m}{2}g(2a), \qquad \text{so} \qquad a^2\dot\theta^2 = u^2 + ga(1 - \cos\theta).$$

Differentiating and cancelling $2a\dot\theta \neq 0$ gives $a\ddot\theta = \frac{1}{2}g\sin\theta$. Then (1) gives

$$F = ma\ddot\theta - mg\sin\theta = -\tfrac{1}{2}mg\sin\theta.$$

So **the frictional force is non-zero** for every $\theta > 0$. (Its magnitude is $\frac{1}{2}mg\sin\theta$, acting back against the direction of motion of O.) Unlike in (i), the rim now carries mass, so the hoop needs a torque about O to make its rotation keep pace with the motion of O, and only friction can supply it.

From (2),

$$R = mg\cos\theta - \frac{m}{a}(u^2 + ga(1 - \cos\theta)) = mg(2\cos\theta - 1) - \frac{mu^2}{a}.$$

At $\theta = 0$, $R = m(g - u^2/a) > 0$ and $F = 0$, so the motion over the edge does begin without slipping. If the hoop kept contact without slipping, R would fall to zero at

$$\cos\theta_1 = \frac{ag + u^2}{2ag} < 1,$$

using $u^2 < ag$. So $\theta_1 > 0$, and there $|F| = \frac{1}{2}mg\sin\theta_1 > 0$.

Ask yourself Why does this force a slip before contact is lost?

With coefficient of friction μ, no slipping requires $|F| \le \mu R$. As $\theta \to \theta_1$ from below, $R \to 0^+$ but $|F| \to \frac{1}{2}mg\sin\theta_1 > 0$, so $|F|/R \to \infty$. Whatever the (finite) value of μ, there is therefore an angle $\theta < \theta_1$, while R is still positive, at which $|F|$ would have to exceed μR. So the hoop **slips before it loses contact** with the table.

Checks

- In (i), if $u^2 \to ag$ the drop tends to 0: the hub is then on the point of flying off at $\theta = 0$, where $g = u^2/a$. If $u \to 0$ the drop is $a/3$, the familiar result for a particle sliding off a smooth sphere starting from rest at the top.
- Stepping the equations numerically with $a = 1\,\text{m}$, $u = 2\,\text{m s}^{-1}$ and $g = 9.81\,\text{m s}^{-2}$ gives a drop of 0.1974 m in (i), matching $(ag - u^2)/(3g)$. In (ii) we found $\cos\theta_1 = 0.7039$, and $|F|/R$ exceeded 10^4 just before $R = 0$.
- Computing the contact force in (ii) directly from the two particles' accelerations ($m\ddot{\mathbf{r}}_G$ minus gravity) reproduces both F and R.

Common slips

- Using rolling kinetic energy $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ about O in (ii) but forgetting that, once pivoting, one particle is fixed at E. It's cleaner to work about E.
- Resolving horizontally and vertically at the edge. At a corner the "normal" direction is along the radius EO, not vertical.
- Concluding "it slips" just because $F \neq 0$. The argument needs $R \to 0$ while $|F|$ stays away from zero.

Beyond STEP

The contrast between the two parts is about the distribution of mass: the same centre-of-mass motion needs different contact forces depending on the moment of inertia, because the angular momentum equation and the linear momentum equation must both be satisfied. In a first mechanics course this becomes rigid-body dynamics, where a body's motion is governed by $\mathbf{F} = m\ddot{\mathbf{r}}_G$ together with the torque equation about G, and contact constraints (rolling, slipping, losing contact) are checked against the inequalities $R \ge 0$ and $|F| \le \mu R$. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers both equations thoroughly.

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